



ENGINEERING MATHEMATICS - II

UE20MA151

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ENGINEERING MATHEMATICS - II

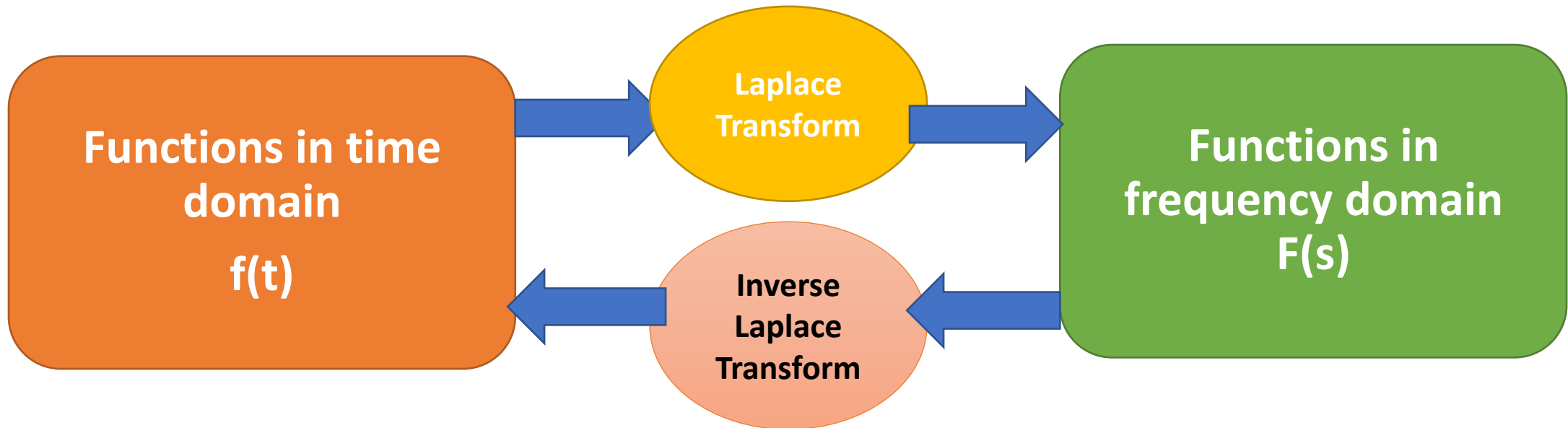


Unit 4 : Inverse Laplace Transform

Session : 1

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Time Domain

$$y^{(2)}(t) + y^{(1)}(t) + y(t) = x(t)$$

$$x(t) = 1$$

$$y(t) = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$$

Frequency Domain

$$H(s) = \frac{1}{s^2 + 3s + 2}$$

$$X(s) = \frac{1}{s}$$

Solve algebraic

$$\frac{1}{s} \cdot \frac{1}{s^2 + 3s + 2}$$

Laplace transform

Inverse Laplace transform

DEFINITION:

If $L\{f(t)\} = F(s)$ then $f(t)$ is known as the Inverse Laplace Transform or Inverse Transform or simply Inverse of $F(s)$ and is denoted by, $L^{-1}\{F(s)\}$.

Therefore, $f(t) = L^{-1}\{F(s)\}$.

Here L^{-1} is known as the Inverse Laplace Transform operator and it satisfies $LL^{-1} = L^{-1}L = 1$.

Note that the Inverse Laplace Transform of $F(s)$ need not exist for all $F(s)$.

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LAPLACE TRANSFORM TABLE

$f(t)$	$F(S)$
1	$\frac{1}{s}, \text{ where } s > 0$
e^{at}	$\frac{1}{s-a}, \text{ where } s > a$
e^{-at}	$\frac{1}{s+a}, \text{ where } s > -a$
$\cosh at$	$\frac{s}{s^2 - a^2}, \text{ where } s > a$
$\sinh at$	$\frac{a}{s^2 - a^2}, \text{ where } s > a$
$\cos at$	$\frac{s}{s^2 + a^2}, \text{ where } s > 0$
$\sin at$	$\frac{a}{s^2 + a^2}, \text{ where } s > 0$

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INVERSE LAPLACE TRANSFORM



1) Find $L^{-1} \left[\frac{s}{s^2+25} \right]$

$$L^{-1} \left[\frac{s}{s^2 + a^2} \right] = \cos at$$

$$L^{-1} \left[\frac{s}{s^2+25} \right] = \cos 5t$$

2) Find $L^{-1} \left[\frac{6}{s^2+4} \right]$

$$L^{-1} \left[\frac{1}{s^2+a^2} \right] = \frac{1}{a} \sin at$$

$$L^{-1} \left[\frac{6}{s^2+2^2} \right] = 6 \cdot \frac{1}{2} \sin 2t = 3 \sin 2t$$

3) Find $L^{-1} \left[\frac{1}{2s-5} \right]$

$$L^{-1} \left[\frac{1}{s-a} \right] = e^{at}$$

$$L^{-1} \left[\frac{1}{2s-5} \right] = L^{-1} \left[\frac{1}{2(s-\frac{5}{2})} \right] = \frac{1}{2} L^{-1} \left[\frac{1}{s-5/2} \right] = \frac{1}{2} e^{5t/2}$$

4) Find $L^{-1} \left[\frac{1}{s^4} \right]$

$$L^{-1} \left[\frac{1}{s^n} \right] = \frac{t^{n-1}}{(n-1)!}, \text{ if } n \text{ is a positive integer}$$
$$= \frac{t^{n-1}}{\Gamma(n)}, \text{ otherwise}$$

$$L^{-1} \left[\frac{1}{s^4} \right] = \frac{t^3}{(3)!} = \frac{t^3}{6}$$

5) Find $L^{-1} \left[\frac{1}{s^{7/2}} \right]$

$$L^{-1} \left[\frac{1}{s^{7/2}} \right] = \frac{t^{5/2}}{\Gamma(\frac{7}{2})} = \frac{8t^{5/2}}{15\sqrt{\pi}}$$

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INVERSE LAPLACE TRANSFORM OF STANDARD FUNCTIONS

Sl No.	$F(S)$	$f(t)$	Example
1.	$L^{-1} \left[\frac{a}{s} \right]$	a	$L^{-1} \left[\frac{5}{s} \right] = 5$
2.	$L^{-1} \left[\frac{1}{s^n} \right]$	$\frac{t^{n-1}}{(n-1)!}$, if n is a positive integer	$L^{-1} \left[\frac{1}{s^4} \right] = \frac{t^3}{3!}$
		$\frac{t^{n-1}}{\Gamma(n)}$	$L^{-1} \left[\frac{1}{s^{7/2}} \right] = \frac{t^{5/2}}{\Gamma(7/2)}$
3.	$L^{-1} \left[\frac{1}{s-a} \right]$	e^{at}	$L^{-1} \left[\frac{1}{s-8} \right] = e^{8t}$
4.	$L^{-1} \left[\frac{1}{s+a} \right]$	e^{-at}	$L^{-1} \left[\frac{1}{s+1} \right] = e^{-t}$

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INVERSE LAPLACE TRANSFORM

Sl No.	$F(S)$	$f(t)$	Example
5.	$L^{-1} \left[\frac{s}{s^2 + a^2} \right]$	$\cos at$	$L^{-1} \left[\frac{s}{s^2 + 25} \right] = \cos 5t$
6.	$L^{-1} \left[\frac{1}{s^2 + a^2} \right]$	$\frac{1}{a} \sin at$	$L^{-1} \left[\frac{1}{s^2 + 16} \right] = \frac{1}{4} \sin 4t$
7.	$L^{-1} \left[\frac{s}{s^2 - a^2} \right]$	$\cosh at$	$L^{-1} \left[\frac{s}{s^2 - 36} \right] = \cosh 6t$
8.	$L^{-1} \left[\frac{1}{s^2 - a^2} \right]$	$\frac{1}{a} \sinh at$	$L^{-1} \left[\frac{1}{s^2 - 9} \right] = \frac{1}{3} \sinh 3t$

Linearity Property :

Let $L\{f(t)\} = F(s)$ & $L\{g(t)\} = G(s)$, and c_1 & c_2 be any two constants then,

$$\begin{aligned} L^{-1}\{c_1 F(s) + c_2 G(s)\} &= c_1 L^{-1}\{F(s)\} + c_2 L^{-1}\{G(s)\} \\ &= c_1 f(t) + c_2 g(t) \end{aligned}$$

1) Find the inverse Laplace Transforms of $\frac{5s-4}{s^2+4}$

Solution : Using Linearity Property,

$$\begin{aligned} L^{-1} \left[\frac{5s-4}{s^2+4} \right] &= 5L^{-1} \left[\frac{s}{s^2+4} \right] - 4L^{-1} \left[\frac{1}{s^2+4} \right] \\ &= 5\cos 2t - 4 \left(\frac{1}{2} \right) \sin 2t \\ &= \mathbf{5\cos 2t - 2\sin 2t} \end{aligned}$$

2) Find the inverse Laplace Transforms of $\frac{3(s^2-2)^2}{2s^5}$

Solution:

$$\begin{aligned} L^{-1} \left[\frac{3(s^2-2)^2}{2s^5} \right] &= \frac{3}{2} L^{-1} \left[\frac{s^4-4s^2+4}{s^5} \right] \\ &= \frac{3}{2} L^{-1} \left[\frac{1}{s} \right] - 6L^{-1} \left[\frac{1}{s^3} \right] + 6L^{-1} \left[\frac{1}{s^5} \right] \\ &= \frac{3}{2} - 6 \frac{t^2}{2!} + 6 \frac{t^4}{4!} \\ &= \frac{3}{2} - 3t^2 + \frac{t^4}{4} \end{aligned}$$

3) Find $L^{-1} \left[\frac{3s-8}{4s^2+5} \right]$

Solution:

$$\begin{aligned} L^{-1} \left[\frac{3s-8}{4s^2+5} \right] &= L^{-1} \left[\frac{3s}{4s^2+5} \right] - L^{-1} \left[\frac{8}{4s^2+5} \right] \\ &= \frac{3}{4} L^{-1} \left[\frac{s}{s^2+\frac{5}{4}} \right] - \frac{8}{4} L^{-1} \left[\frac{1}{s^2+\frac{5}{4}} \right] \\ &= \frac{3}{4} \cos \frac{\sqrt{5} t}{2} - \frac{8}{4} \frac{2}{\sqrt{5}} \sin \frac{\sqrt{5} t}{2} \\ &= \frac{3}{4} \cos \frac{\sqrt{5} t}{2} - \frac{4}{\sqrt{5}} \sin \frac{\sqrt{5} t}{2} \end{aligned}$$

4) Find $L^{-1} \left[\frac{2s-5}{9s^2-25} \right]$

Solution:

$$\begin{aligned} L^{-1} \left[\frac{2s-5}{9s^2-25} \right] &= L^{-1} \left[\frac{2s}{9s^2-25} \right] - L^{-1} \left[\frac{5}{9s^2-25} \right] \\ &= \frac{2}{9} L^{-1} \left[\frac{s}{s^2-\frac{25}{9}} \right] - \frac{5}{9} L^{-1} \left[\frac{1}{s^2-\frac{25}{9}} \right] \\ &= \frac{2}{9} \cosh \frac{5t}{3} - \frac{5}{9} \frac{3}{5} \sinh \frac{5t}{3} \\ &= \frac{2}{9} \cosh \frac{5t}{3} - \frac{1}{3} \sinh \frac{5t}{3} \end{aligned}$$



THANK YOU

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